Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	**Slot:** L43 + L44
Regno: 23BCE1145	**Name:** S Vishwajith

EX8 – SHA 512

Code:

```
import copy

Buffers = [
0x6A09E667F3BCC908, 0xBB67AE8584CAA73B, 0x3C6EF372FE94F82B, 0xA54FF53A5F1D36F1,
0x510E527FADE682D1, 0x9B05688C2B3E6C1F, 0x1F83D9ABFB41BD6B, 0x5BE0CD19137E2179
]

Keys = [
0x428A2F98D728AE22, 0x7137449123EF65CD, 0xB5C0FBCFEC4D3B2F, 0xE9B5DBA58189DBBC,
0x3956C25BF348B538, 0x59F111F1B605D019, 0x923F82A4AF194F9B, 0xAB1C5ED5DA6D8118,
0xD807AA98A3030242, 0x12835B0145706FBE, 0x243185BE4EE4B28C, 0x550C7DC3D5FFB4E2,
0x72BE5D74F27B896F, 0x80DEB1FE3B1696B1, 0x9BDC06A725C71235, 0xC19BF174CF692694,
0xE49B69C19EF14AD2, 0xEFBE4786384F25E3, 0x0FC19DC68B8CD5B5, 0x240CA1CC77AC9C65,
0x2DE92C6F592B0275, 0x4A7484AA6EA6E483, 0x5CB0A9DCBD41FBD4, 0x76F988DA831153B5,
0x983E5152EE66DFAB, 0xA831C66D2DB43210, 0xB00327C898FB213F, 0xBF597FC7BEEF0EE4,
0xC6E00BF33DA88FC2, 0xD5A79147930AA725, 0x06CA6351E003826F, 0x142929670A0E6E70,
0x27B70A8546D22FFC, 0x2E1B21385C26C926, 0x4D2C6DFC5AC42AED, 0x53380D139D95B3DF,
0x650A73548BAF63DE, 0x766A0ABB3C77B2A8, 0x81C2C92E47EDAEE6, 0x92722C851482353B,
0xA2BFE8A14CF10364, 0xA81A664BBC423001, 0xC24B8B70D0F89791, 0xC76C51A30654BE30,
0xD192E819D6EF5218, 0xD69906245565A910, 0xF40E35855771202A, 0x106AA07032BBD1B8,
0x19A4C116B8D2D0C8, 0x1E376C085141AB53, 0x2748774CDF8EEB99, 0x34B0BCB5E19B48A8,
0x391C0CB3C5C95A63, 0x4ED8AA4AE3418ACB, 0x5B9CCA4F7763E373, 0x682E6FF3D6B2B8A3,
0x748F82EE5DEFB2FC, 0x78A5636F43172F60, 0x84C87814A1F0AB72, 0x8CC702081A6439EC,
```

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0x90BEFFFA23631E28, 0xA4506CEBDE82BDE9, 0xBEF9A3F7B2C67915, 0xC67178F2E372532B,
0xCA273ECEEA26619C, 0xD186B8C721C0C207, 0xEADA7DD6CDE0EB1E, 0xF57D4F7FEE6ED178,
0x06F067AA72176FBA, 0x0A637DC5A2C898A6, 0x113F9804BEF90DAE, 0x1B710B35131C471B,
0x28DB77F523047D84, 0x32CAAB7B40C72493, 0x3C9EBE0A15C9BEBC, 0x431D67C49C100D4C,
0x4CC5D4BECB3E42B6, 0x597F299CFC657E2A, 0x5FCB6FAB3AD6FAEC, 0x6C44198C4A475817,
]

state = copy.deepcopy(Buffers)

def convert_hex_to_int(h):
    if(h=='''): return 0
    else: return int(h,16)

def number_of_blocks(l):
    n = (-l-1-128)%1024
    if(n==0): n=1024
    return((l+128+n+1)//1024, n)

def rotate_right(w,n):
    w=bin(w)[2:].rjust(64,'0')
    w=w[-n:]+w[:-n]
    return int(w,2)

def shift_512(w,n):
    nums=[(1,8,7),(19,61,6)]
    x=nums[n]
    w=convert_hex_to_int(w)
    return (rotate_right(w,x[0]) ^ rotate_right(w,x[1]) ^ w>>x[2])

def sigma(w,n):
    nums=[(28,34,39),(14,18,41)]
```

```
    x=nums[n]
    return (rotate_right(w,x[0]) ^ rotate_right(w,x[1]) ^ rotate_right(w,x[2]))

def ch(e,f,g):
    return (e & f) ^ ((e ^ (2**64-1)) & g)

def maj(a,b,c):
    return (a & b) ^ (b & c) ^ (c & a)

def generate_words(msg):
    w = []
    for i in range(16):
        w.append(msg[16*i:16*i+16])
    for i in range(16,80):
        nw = ((shift_512(w[i-2],1)+shift_512(w[i-15],0)+convert_hex_to_int(w[i-
16])+convert_hex_to_int(w[i-7]))%(2**64))
        w.append(hex(nw)[2:])
    return w

def sha(msg):
    [a,b,c,d,e,f,g,h]=state
    words = generate_words(hex(msg)[2:].rjust(256,'0'))
    for r in range(80):
        t1 = (h+ch(e,f,g)+sigma(e,1)+convert_hex_to_int(words[r])+Keys[r])%(2**64)
        t2 = (sigma(a,0)+maj(a,b,c))%(2**64)
        nh = g
        ng = f
        nf = e
        ne = (d+t1)%(2**64)
        nd = c
        nc = b
        nb = a
```

```
        na = (t1+t2)%(2**64)
        a,b,c,d,e,f,g,h = na,nb,nc,nd,ne,nf,ng,nh
        string_output = "0x"
        for x in [a,b,c,d,e,f,g,h]:
            string_output+=hex(x)[2:].rjust(16,'0')
        print(f"Round {r+1}: {string_output}")
    bufs = [a,b,c,d,e,f,g,h]
    string_output = "0x"
    for i in range(8):
        state[i] = (state[i] + bufs[i])%(2**64)
        string_output+=hex(state[i])[2:].rjust(16,'0')
    print(f"Hash value: {string_output}")

s = input("Enter the string here: ")
h = s.encode('ascii').hex()
l = 4*len(h)
print(f"\nInput bits: {l}")
h = (convert_hex_to_int(h)<<1) + 1
nblocks, npadding = number_of_blocks(l)
h = (h<<(npadding+128))+(l)
print(f"Number of blocks: {nblocks}")
print("Input blocks: 0x",end="")

blocks = []
for i in range(nblocks):
    blocks.append(convert_hex_to_int(hex(h)[2+256*i:258+256*i].rjust(256,'0')))
    print(hex(h)[2+256*i:258+256*i].rjust(256,'0'), end=" ")
print('\n')

for block in blocks: sha(block)
```

```
string_output = "0x"

for i in range(8):

    string_output+=hex(state[i])[2:].rjust(16,'0')

print(f"Final Hash: {string_output}")
```

Output:

Test Case 1:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-10> python .\SHA_512.py
Enter the string here: 23BCE1145

Input bits: 72
Number of blocks: 1
Input blocks: 0x323342434531313435800000000000
```

Test Case 2:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-10> python .\SHA_512.py
Enter the string here: BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145BCE1145

Input bits: 896
Number of blocks: 2
Input blocks: 0x4243453131343542434531313435424345313134354243453131343542434531313435424345313134354243453131343542434531313435424345313134354
```

```
Round 1: 0x94615ec0e49f1625e7d49e61dd62bb2bd4809a684bcd1833fcb4b7ab469e04c139e9136b7e28691ad04e35b170de8ec3297e59015d94c520f43f95dd79de7050
Round 2: 0x4746e9d7c9cac90794615ec0e49f1625e7d49e61dd62bb2bd4809a684bcd18337e1b6925da743aa239e9136b7e28691ad04e35b170de8ec3297e59015d94c520
Round 3: 0x96eb5ee26d6a058c4746e9d7c9cac90794615ec0e49f1625e7d49e61dd62bb2be2d962e1403ad7be7e1b6925da743aa239e9136b7e28691ad04e35b170de8ec3
Round 4: 0xf0751d9ebcd1d59d96eb5ee26d6a058c4746e9d7c9cac90794615ec0e49f1625b8c64f9c5a479ca0e2d962e1403ad7be7e1b6925da743aa239e9136b7e28691a
Round 5: 0xeb59143a5de52be0f0751d9ebcd1d59d96eb5ee26d6a058c4746e9d7c9cac9074a028611140d07c1b8c64f9c5a479ca0e2d962e1403ad7be7e1b6925da743aa2
Round 6: 0x0770b96d0689adfceb59143a5de52be0f0751d9ebcd1d59d96eb5ee26d6a058c1eae47e903e1bcf44a028611140d07c1b8c64f9c5a479ca0e2d962e1403ad7be
Round 7: 0x5e6e468ae7474c6b0770b96d0689adfceb59143a5de52be0f0751d9ebcd1d59d1eb2e04aa9164a021eae47e903e1bcf44a028611140d07c1b8c64f9c5a479ca0
Round 8: 0x8a451a6fb17dd64e5e6e468ae7474c6b0770b96d0689adfceb59143a5de52be052d688ac3ec758821eb2e04aa9164a021eae47e903e1bcf44a028611140d07c1
Round 9: 0x5640b97de50bcd238a451a6fb17dd64e5e6e468ae7474c6b0770b96d0689adfc0e54c870ffcec24152d688ac3ec758821eb2e04aa9164a021eae47e903e1bcf4
Round 10: 0xf68c3a13980864be5640b97de50bcd238a451a6fb17dd64e5e6e468ae7474c6bcd84a228c20b0d5c0e54c870ffcec24152d688ac3ec758821eb2e04aa9164a02
Round 11: 0x5b786c2c5eb0c446f68c3a13980864be5640b97de50bcd238a451a6fb17dd64ea1ee362dec6094bacd84a228c20b0d5c0e54c870ffcec24152d688ac3ec75882
Round 12: 0x930f5dac7ab87e7a5b786c2c5eb0c446f68c3a13980864be5640b97de50bcd2322ef2ae1a23f678ca1ee362dec6094bacd84a228c20b0d5c0e54c870ffcec241
Round 13: 0x66ed69bc35626804930f5dac7ab87e7a5b786c2c5eb0c446f68c3a13980864befc451e425ea5bd9222ef2ae1a23f678ca1ee362dec6094bacd84a228c20b0d5c
Round 14: 0x9ed4e7b495bf212a66ed69bc35626804930f5dac7ab87e7a5b786c2c5eb0c4461eded5842f76fe23fc451e425ea5bd9222ef2ae1a23f678ca1ee362dec6094ba
Round 15: 0xadc69547e8d5917e9ed4e7b495bf212a66ed69bc35626804930f5dac7ab87e7a5abaef16f205022f1eded5842f76fe23fc451e425ea5bd9222ef2ae1a23f678c
Round 16: 0x0e3d6eaaf0bb1868adc69547e8d5917e9ed4e7b495bf212a66ed69bc35626804f9ca0e0b2b38f9af5abaef16f205022f1eded5842f76fe23fc451e425ea5bd92
Round 17: 0x8e140b01b1346eff0e3d6eaaf0bb1868adc69547e8d5917e9ed4e7b495bf212a83ae566cc591f9c1f9ca0e0b2b38f9af5abaef16f205022f1eded5842f76fe23
Round 18: 0xa23a1e71b653df168e140b01b1346eff0e3d6eaaf0bb1868adc69547e8d5917e36915a875d844cf583ae566cc591f9c1f9ca0e0b2b38f9af5abaef16f205022f
Round 19: 0x138fe7e482112ec5a23a1e71b653df168e140b01b1346eff0e3d6eaaf0bb18684754cdf914484d0a36915a875d844cf583ae566cc591f9c1f9ca0e0b2b38f9af
Round 20: 0xcb23482ba27e387f138fe7e482112ec5a23a1e71b653df168e140b01b1346eff8eb021b6eaed49924754cdf914484d0a36915a875d844cf583ae566cc591f9c1
Round 21: 0xb269124addb3e732cb23482ba27e387f138fe7e482112ec5a23a1e71b653df162518d9572d2b01ba8eb021b6eaed49924754cdf914484d0a36915a875d844cf5
Round 22: 0xdd7f9d9a3c5a16a6b269124addb3e732cb23482ba27e387f138fe7e482112ec5f2a48c77810b152a2518d9572d2b01ba8eb021b6eaed49924754cdf914484d0a
Round 23: 0xd4728952aecb4474dd7f9d9a3c5a16a6b269124addb3e732cb23482ba27e387f8dc7fd4347457b05f2a48c77810b152a2518d9572d2b01ba8eb021b6eaed4992
Round 24: 0xa891c7ef7914f6ebd4728952aecb4474dd7f9d9a3c5a16a6b269124addb3e732bcde75e7c31b16c88dc7fd4347457b05f2a48c77810b152a2518d9572d2b01ba
Round 25: 0x68fb828a7e0f439ca891c7ef7914f6ebd4728952aecb4474dd7f9d9a3c5a16a6acf88c0d6affa765bcde75e7c31b16c88dc7fd4347457b05f2a48c77810b152a
Round 26: 0xd111b1acc9e518e768fb828a7e0f439ca891c7ef7914f6ebd4728952aecb44745a11d3d5dad8a991acf88c0d6affa765bcde75e7c31b16c88dc7fd4347457b05
Round 27: 0x2fc2a218c330b17dd111b1acc9e518e768fb828a7e0f439ca891c7ef7914f6eb25e8a547cdf94d3e5a11d3d5dad8a991acf88c0d6affa765bcde75e7c31b16c8
Round 28: 0xccb045f3b7cf2e202fc2a218c330b17dd111b1acc9e518e768fb828a7e0f439cd92d106b183a06c025e8a547cdf94d3e5a11d3d5dad8a991acf88c0d6affa765
Round 29: 0xda500727372889d7ccb045f3b7cf2e202fc2a218c330b17dd111b1acc9e518e7ff4c302037a16642d92d106b183a06c025e8a547cdf94d3e5a11d3d5dad8a991
Round 30: 0xda4a58c2537847efda500727372889d7ccb045f3b7cf2e202fc2a218c330b17deb5f151f6a13e3adff4c302037a16642d92d106b183a06c025e8a547cdf94d3e
Round 31: 0x3c55dcc0f0b9e3c4da4a58c2537847efda500727372889d7ccb045f3b7cf2e2069cb996ceefef6b6eb5f151f6a13e3adff4c302037a16642d92d106b183a06c0
Round 32: 0xcbbf08f4214517da3c55dcc0f0b9e3c4da4a58c2537847efda500727372889d707cdbab023ab71ad69cb996ceefef6b6eb5f151f6a13e3adff4c302037a16642
Round 33: 0xb78b28888fb1c430cbbf08f4214517da3c55dcc0f0b9e3c4da4a58c2537847efc339247998d8d2cb07cdbab023ab71ad69cb996ceefef6b6eb5f151f6a13e3ad
Round 34: 0xad839b69a232b358b78b28888fb1c430cbbf08f4214517da3c55dcc0f0b9e3c4fe3f99dc1fb9de7cc339247998d8d2cb07cdbab023ab71ad69cb996ceefef6b6
Round 35: 0xec2a5df8a32fa039ad839b69a232b358b78b28888fb1c430cbbf08f4214517dae0113596aeb5952ffe3f99dc1fb9de7cc339247998d8d2cb07cdbab023ab71ad
Round 36: 0x45ca7af4c923df7fec2a5df8a32fa039ad839b69a232b358b78b28888fb1c4307886b52f7cbe3ad2e0113596aeb5952ffe3f99dc1fb9de7cc339247998d8d2cb
Round 37: 0xeb7ac317ca65c40645ca7af4c923df7fec2a5df8a32fa039ad839b69a232b3586b94481243952fb97886b52f7cbe3ad2e0113596aeb5952ffe3f99dc1fb9de7c
Round 38: 0xf58676ce46c9a520eb7ac317ca65c40645ca7af4c923df7fec2a5df8a32fa0393a4dbb5c1d301ffa6b94481243952fb97886b52f7cbe3ad2e0113596aeb5952f
Round 39: 0x406094466b9790b4f58676ce46c9a520eb7ac317ca65c40645ca7af4c923df7fe51ad75b1f776e0d3a4dbb5c1d301ffa6b94481243952fb97886b52f7cbe3ad2
Round 40: 0xc0ebe429c606694a406094466b9790b4f58676ce46c9a520eb7ac317ca65c4067607c88fc5d32446e51ad75b1f776e0d3a4dbb5c1d301ffa6b94481243952fb9
Round 41: 0x776b9431717b5023c0ebe429c606694a406094466b9790b4f58676ce46c9a5205f9860ceca5406ab7607c88fc5d32446e51ad75b1f776e0d3a4dbb5c1d301ffa
Round 42: 0x7a6be7e988139830776b9431717b5023c0ebe429c606694a406094466b9790b4067da9f67e387bfb5f9860ceca5406ab7607c88fc5d32446e51ad75b1f776e0d
Round 43: 0x4cac09f7ddd1fa837a6be7e988139830776b9431717b5023c0ebe429c606694aea7377d8b085e268067da9f67e387bfb5f9860ceca5406ab7607c88fc5d32446
Round 44: 0xe6c6bb1f9c2a63844cac09f7ddd1fa837a6be7e988139830776b9431717b50236376a3f6696a2f98ea7377d8b085e268067da9f67e387bfb5f9860ceca5406ab
Round 45: 0xbfa0db0935256756e6c6bb1f9c2a63844cac09f7ddd1fa83ac0ff335c25d52bead88ced2ae752ab16376a3f6696a2f98ea7377d8b085e268
Round 46: 0x2dbe0cdcd005e5fbbfa0db0935256756e6c6bb1f9c2a63844cac09f7ddd1fa83ac0ff335c25d52bead88ced2ae752ab16376a3f6696a2f98ea7377d8b085e268
Round 47: 0x90848045e3a729c42dbe0cdcd005e5fbbfa0db0935256756e6c6bb1f9c2a63849f8ac218467f898eac0ff335c25d52bead88ced2ae752ab16376a3f6696a2f98
Round 48: 0xff5fff33953a0c8090848045e3a729c42dbe0cdcd005e5fbbfa0db0935256756 5f2619999af6dcdf9f8ac218467f898eac0ff335c25d52bead88ced2ae752ab1
Round 49: 0x43e217d225eccd63ff5fff33953a0c8090848045e3a729c42dbe0cdcd005e5fb7420274ebe1453ee5f2619999af6dcdf9f8ac218467f898eac0ff335c25d52be
Round 50: 0x9774d3c99e98c55443e217d225eccd63ff5fff33953a0c8090848045e3a729c47d700c5ca0926096742027 4ebe1453ee5f2619999af6dcdf9f8ac218467f898e
Round 51: 0xd8a3b25cf0c9bc589774d3c99e98c55443e217d225eccd63ff5fff33953a0c80749a053842f937107d700c5ca0926096742027 4ebe1453ee5f2619999af6dcdf
Round 52: 0xb8bd9f8d24bd3edad8a3b25cf0c9bc589774d3c99e98c55443e217d225eccd635af2d104f78befce749a053842f937107d700c5ca0926096742027 4ebe1453ee
Round 53: 0x64531baaa6aa05e1b8bd9f8d24bd3edad8a3b25cf0c9bc589774d3c99e98c554f6c9c74e877532de5af2d104f78befce749a053842f937107d700c5ca0926096
Round 54: 0x8eec841aee9a2fa064531baaa6aa05e1b8bd9f8d24bd3edad8a3b25cf0c9bc589776630f2099bdda5f6c9c74e877532de5af2d104f78befce749a053842f93710
Round 55: 0x1b65ad265693944a8eec841aee9a2fa064531baaa6aa05e1b8bd9f8d24bd3edad327a39143467229766 30f2099bdda5f6c9c74e877532de5af2d104f78befce
Round 56: 0x2800fc085719a9f81b65ad265693944a8eec841aee9a2fa064531baaa6aa05e1222887cef087707cdf327a39143467229766 30f2099bdda5f6c9c74e877532de
Round 57: 0x98a2e613988c70472800fc085719a9f81b65ad265693944a8eec841aee9a2fa07d9e8f046e3ac9c1222887cef087707cdf327a39143467229766 30f2099bdda5
Round 58: 0x7875551c73f19d5b98a2e613988c70472800fc085719a9f81b65ad265693944aa56becac28ba7f407d9e8f046e3ac9c1222887cef087707cdf327a3914346722
Round 59: 0xc25ac521caef3b637875551c73f19d5b98a2e613988c70472800fc085719a9f88a9d2c53de302828a56becac28ba7f407d9e8f046e3ac9c1222887cef087707c
Round 60: 0x5bdfbdd8868ff47fc25ac521caef3b637875551c73f19d5b98a2e613988c70470b904e3ceffe0e568a9d2c53de302828a56becac28ba7f407d9e8f046e3ac9c1
Round 61: 0xadcb335369098caa5bdfbdd8868ff47fc25ac521caef3b637875551c73f19d5bf3bc9f4ba8eab1f10b904e3ceffe0e568a9d2c53de302828a56becac28ba7f40
Round 62: 0x51731ffe6c31bc7fadcb335369098caa5bdfbdd8868ff47fc25ac521caef3b631204052ff2f92debf3bc9f4ba8eab1f10b904e3ceffe0e568a9d2c53de302828
Round 63: 0x4777ffc52cf0e94e51731ffe6c31bc7fadcb335369098caa5bdfbdd8868ff47f0b27b97a4e068a0c1204052ff2f92debf3bc9f4ba8eab1f10b904e3ceffe0e56
Round 64: 0x8ae92df55bd720ae4777ffc52cf0e94e51731ffe6c31bc7fadcb335369098caa92ea5c90fc087c630b27b97a4e068a0c1204052ff2f92debf3bc9f4ba8eab1f1
Round 65: 0xa7753daa7dd7db148ae92df55bd720ae4777ffc52cf0e94e51731ffe6c31bc7f109cc48628b278dd92ea5c90fc087c630b27b97a4e068a0c1204052ff2f92deb
Round 66: 0xe89dad8fa189b280a7753daa7dd7db148ae92df55bd720ae4777ffc52cf0e94e9ba7a76f1823063b109cc48628b278dd92ea5c90fc087c630b27b97a4e068a0c
Round 67: 0xf2a8395a28e1385ce89dad8fa189b280a7753daa7dd7db148ae92df55bd720ae9f68b3a429cde6649ba7a76f1823063b109cc48628b278dd92ea5c90fc087c63
Round 68: 0x33763fbade52bf86f2a8395a28e1385ce89dad8fa189b280a7753daa7dd7db142b28267d609b5a9a9f68b3a429cde6649ba7a76f1823063b109cc48628b278dd
Round 69: 0xe980497c2dff1ec833763fbade52bf86f2a8395a28e1385ce89dad8fa189b2807735568e4b3ca71c2b28267d609b5a9a9f68b3a429cde6649ba7a76f1823063b
Round 70: 0x0c958b87d8ff3655e980497c2dff1ec833763fbade52bf86f2a8395a28e1385cd4bd2a9c19de24fa7735568e4b3ca71c2b28267d609b5a9a9f68b3a429cde664
Round 71: 0xd8753cbc44bb08cf0c958b87d8ff3655e980497c2dff1ec833763fbade52bf862b0c13ae703810dcd4bd2a9c19de24fa7735568e4b3ca71c2b28267d609b5a9a
Round 72: 0xe23f29d4c6bc3067d8753cbc44bb08cf0c958b87d8ff3655e980497c2dff1ec82b0926f36911628e2b0c13ae703810dcd4bd2a9c19de24fa7735568e4b3ca71c
Round 73: 0x1c742304acd698ebe23f29d4c6bc3067d8753cbc44bb08cf0c958b87d8ff365549ddd0e5015d5e022b0926f36911628e2b0c13ae703810dcd4bd2a9c19de24fa
Round 74: 0xfe9df9069858104d1c742304acd698ebe23f29d4c6bc3067d8753cbc44bb08cf43b8d53ed6315ce649ddd0e5015d5e022b0926f36911628e2b0c13ae703810dc
Round 75: 0x20793cbf2a04b5fcfe9df9069858104d1c742304acd698ebe23f29d4c6bc3067cc099f8ad45dd88243b8d53ed6315ce649ddd0e5015d5e022b0926f36911628e
Round 76: 0x8859de28aad62fe220793cbf2a04b5fcfe9df9069858104d1c742304acd698eb197d531fd17058b8cc099f8ad45dd88243b8d53ed6315ce649ddd0e5015d5e02
Round 77: 0x866d8d8827c848628859de28aad62fe220793cbf2a04b5fcfe9df9069858104d2404dace47af84a5197d531fd17058b8cc099f8ad45dd88243b8d53ed6315ce6
Round 78: 0x2efccea2907174d0866d8d8827c848628859de28aad62fe220793cbf2a04b5fc47e82660fbb5f6402404dace47af84a5197d531fd17058b8cc099f8ad45dd882
Round 79: 0xf6fbd6a785ab02282efccea2907174d0866d8d8827c848628859de28aad62fe2a2cb51dec8b12b2647e82660fbb5f6402404dace47af84a5197d531fd17058b8
Round 80: 0x2e5ca25e257bee2bf6fbd6a785ab02282efccea2907174d0866d8d8827c848621ae6fde3467a7531a2cb51dec8b12b2647e82660fbb5f6402404dace47af84a5
Hash value: 0x163140c002dea956cb7c710fd1781a5b2bb1864dd70f79918353b99ee4931c33eb353394b75903f4cc49aae02645f0463c27bc3e75946690b9f626d09e0dbf65
Final Hash: 0x163140c002dea956cb7c710fd1781a5b2bb1864dd70f79918353b99ee4931c33eb353394b75903f4cc49aae02645f0463c27bc3e75946690b9f626d09e0dbf65
```

Test Case 3:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-10> python .\SHA_512.py
Enter the string here: 23BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE114523BCE11
4523BCE1145

Input bits: 1152
Number of blocks: 2
Input blocks: 0x3233424345313134353233424345313134353233424345313134353233424345313134353233424345313134353233424345313134353233424345313
3435323334243453131343532334243453131343532334243453131343532334243453131343532334243453131343532334243453131343532334243453131343532
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000480

Round 1: 0xc780ad7c022e0f296a09e667f3bcc908bb67ae8584caa73b3c6ef372fe94f82b299be0f7bfe650c5510e527fade682d19b05688c2b3e6c1f1f83d9abfb41bd6b
Round 2: 0xcb6694ce0700ebc2c780ad7c022e0f296a09e667f3bcc908bb67ae8584caa73b4264ac93288156ff299be0f7bfe650c5510e527fade682d19b05688c2b3e6c1f
Round 3: 0x169e4ea972d207b9cb6694ce0700ebc2c780ad7c022e0f296a09e667f3bcc9089944d1b57349bef24264ac93288156ff299be0f7bfe650c5510e527fade682d1
Round 4: 0xc69f57247bbe81ae169e4ea972d207b9cb6694ce0700ebc2c780ad7c022e0f2945b152300f0af56d9944d1b57349bef24264ac93288156ff299be0f7bfe650c5
Round 5: 0xbce1b540ad4d033cc69f57247bbe81ae169e4ea972d207b9cb6694ce0700ebc2c7d3448bb608f59b45b152300f0af56d9944d1b57349bef24264ac93288156ff
Round 6: 0xfb68c9d37e2833b2bce1b540ad4d033cc69f57247bbe81ae169e4ea972d207b9b1561e7e70f02188cfd3448bb608f59b45b152300f0af56d9944d1b57349bef2
Round 7: 0x81149797bcaeae2efb68c9d37e2833b2bce1b540ad4d033cc69f57247bbe81aefcd3a95c42b43359b1561e7e70f02188cfd3448bb608f59b45b152300f0af56d
Round 8: 0x1a0cc38e9b789dd981149797bcaeae2efb68c9d37e2833b2bce1b540ad4d033c1c93709f6efc8698fcd3a95c42b43359b1561e7e70f02188cfd3448bb608f59b
Round 9: 0xb4d33b8441e94a661a0cc38e9b789dd981149797bcaeae2efb68c9d37e2833b2ca472150dc927c5b1c93709f6efc8698fcd3a95c42b43359b1561e7e70f02188
Round 10: 0x8f39b07f728f8fcfb4d33b8441e94a661a0cc38e9b789dd981149797bcaeae2ef420c06629606441ca472150dc927c5b1c93709f6efc8698fcd3a95c42b43359
Round 11: 0x82a56492182ae5ff8f39b07f728f8fcfb4d33b8441e94a661a0cc38e9b789dd95ae24886f1a4ebd6f420c06629606441ca472150dc927c5b1c93709f6efc8698
Round 12: 0xf4149513962f2b8d82a56492182ae5ff8f39b07f728f8fcfb4d33b8441e94a6666db74bfab019eab5ae24886f1a4ebd6f420c06629606441ca472150dc927c5b
Round 13: 0xaa7a175137e76c73f4149513962f2b8d82a56492182ae5ff8f39b07f728f8fcf38c272573ffca9b366db74bfab019eab5ae24886f1a4ebd6f420c06629606441
Round 14: 0xe467dab0ce9acc38aa7a175137e76c73f4149513962f2b8d82a56492182ae5ff3f8ca31e808fe91f38c272573ffca9b366db74bfab019eab5ae24886f1a4ebd6
Round 15: 0x264c496848a9539be467dab0ce9acc38aa7a175137e76c73f4149513962f2b8d08e3d1d0bb783b4c3f8ca31e808fe91f38c272573ffca9b366db74bfab019eab
Round 16: 0x9e98738b3225bc8e264c496848a9539be467dab0ce9acc38aa7a175137e76c73a4106d3ade8e7f9408e3d1d0bb783b4c3f8ca31e808fe91f38c272573ffca9b3
Round 17: 0x53e8c0424de2f53c9e98738b3225bc8e264c496848a9539be467dab0ce9acc385f5f8f37e804705fa4106d3ade8e7f9408e3d1d0bb783b4c3f8ca31e808fe91f
Round 18: 0x2e3c7fce5f5e88cc53e8c0424de2f53c9e98738b3225bc8e264c496848a9539b320fedd4928579b25f5f8f37e804705fa4106d3ade8e7f9408e3d1d0bb783b4c
Round 19: 0xae478f2895b7d4132e3c7fce5f5e88cc53e8c0424de2f53c9e98738b3225bc8eb759d0e60ee5cf39320fedd4928579b25f5f8f37e804705fa4106d3ade8e7f94
Round 20: 0xd8c60cf772090721ae478f2895b7d4132e3c7fce5f5e88cc53e8c0424de2f53c19b5d5a21d825136b759d0e60ee5cf39320fedd4928579b25f5f8f37e804705f
Round 21: 0x6f8036ef48853491d8c60cf772090721ae478f2895b7d4132e3c7fce5f5e88cc022cc5678567e57919b5d5a21d825136b759d0e60ee5cf39320fedd4928579b2
Round 22: 0x7aa196ee5b71e4e16f8036ef48853491d8c60cf772090721ae478f2895b7d4133534f971e90ef67a022cc5678567e57919b5d5a21d825136b759d0e60ee5cf39
Round 23: 0x28308fb22d2368987aa196ee5b71e4e16f8036ef48853491d8c60cf7720907215e8333053295cf973534f971e90ef67a022cc5678567e57919b5d5a21d825136
Round 24: 0x7de7f640eafc191428308fb22d2368987aa196ee5b71e4e16f8036ef48853491aee956d45a8f01d35e8333053295cf973534f971e90ef67a022cc5678567e579
Round 25: 0xa24c402d2ff936d37de7f640eafc191428308fb22d2368987aa196ee5b71e4e1847f7109230fac62aee956d45a8f01d35e8333053295cf973534f971e90ef67a
Round 26: 0x229ca6f31226a641a24c402d2ff936d37de7f640eafc191428308fb22d23689886a6a4ddd28bc839847f7109230fac62aee956d45a8f01d35e8333053295cf97
Round 27: 0xb39ff98436cdc862229ca6f31226a641a24c402d2ff936d37de7f640eafc1914b83c220655cc7ae986a6a4ddd28bc839847f7109230fac62aee956d45a8f01d3
Round 28: 0xb6015887c5725f06b39ff98436cdc862229ca6f31226a641a24c402d2ff936d32849fe8029017806b83c220655cc7ae986a6a4ddd28bc839847f7109230fac62
Round 29: 0xc5291eb9d8f520fcb6015887c5725f06b39ff98436cdc862229ca6f31226a6410ae3e34b318daca12849fe8029017806b83c220655cc7ae986a6a4ddd28bc839
Round 30: 0xf715568cbafbe5bac5291eb9d8f520fcb6015887c5725f06b39ff98436cdc862d7c9f7475ddb45870ae3e34b318daca12849fe8029017806b83c220655cc7ae9
Round 31: 0x2cc2e29989742d61f715568cbafbe5bac5291eb9d8f520fcb6015887c5725f0650ec28852dd07840d7c9f7475ddb45870ae3e34b318daca12849fe8029017806
Round 32: 0x89de5cb652d2674b2cc2e29989742d61f715568cbafbe5bac5291eb9d8f520fc94d129943a088b5e50ec28852dd07840d7c9f7475ddb45870ae3e34b318daca1
Round 33: 0x586df8436681327a89de5cb652d2674b2cc2e29989742d61f715568cbafbe5ba9a8976bbbc020d1a94d129943a088b5e50ec28852dd07840d7c9f7475ddb4587
Round 34: 0x7ded4db47f48402a586df8436681327a89de5cb652d2674b2cc2e29989742d6134b5424e12ffdca39a8976bbbc020d1a94d129943a088b5e50ec28852dd07840
Round 35: 0x9b4eb4284e2a212d7ded4db47f48402a586df8436681327a89de5cb652d2674bea7bb7de4b89997334b5424e12ffdca39a8976bbbc020d1a94d129943a088b5e
Round 36: 0x0b845d55e4eae3579b4eb4284e2a212d7ded4db47f48402a586df8436681327a9a3fef1b40a6b792ea7bb7de4b89997334b5424e12ffdca39a8976bbbc020d1a
Round 37: 0x9a4431d34f409c010b845d55e4eae3579b4eb4284e2a212d7ded4db47f48402a3a88747e72a3a29b9a3fef1b40a6b792ea7bb7de4b89997334b5424e12ffdca3
Round 38: 0xfb881d1304ebdb059a4431d34f409c010b845d55e4eae3579b4eb4284e2a212d5ce9cf5473e2e1073a88747e72a3a29b9a3fef1b40a6b792ea7bb7de4b899973
Round 39: 0x9d3e3716b0aec0a8fb881d1304ebdb059a4431d34f409c010b845d55e4eae357f37a3cc2ec5940825ce9cf5473e2e1073a88747e72a3a29b9a3fef1b40a6b792
Round 40: 0x251481095f08ec229d3e3716b0aec0a8fb881d1304ebdb059a4431d34f409c0109e5c229997276b8f37a3cc2ec5940825ce9cf5473e2e1073a88747e72a3a29b
Round 41: 0xc811dba59d7bd243251481095f08ec229d3e3716b0aec0a8fb881d1304ebdb05ef470f733bca8c4f09e5c229997276b8f37a3cc2ec5940825ce9cf5473e2e107
Round 42: 0xf39e54755bbbdbc8c811dba59d7bd243251481095f08ec229d3e3716b0aec0a86638b779cef307a4ef470f733bca8c4f09e5c229997276b8f37a3cc2ec594082
Round 43: 0xfa5a5bfc6673a76af39e54755bbbdbc8c811dba59d7bd243251481095f08ec22ae9f842323009ef06638b779cef307a4ef470f733bca8c4f09e5c229997276b8
Round 44: 0xf8065e6147f7d8e5fa5a5bfc6673a76af39e54755bbbdbc8c811dba59d7bd2439c960b3978bd542fae9f842323009ef06638b779cef307a4ef470f733bca8c4f
Round 45: 0xad345b88182bae52f8065e6147f7d8e5fa5a5bfc6673a76af39e54755bbbdbc88f1840e1381f277d9c960b3978bd542fae9f842323009ef06638b779cef307a4
Round 46: 0x7d4977903ab3681cad345b88182bae52f8065e6147f7d8e5fa5a5bfc6673a76ae451163a72a8ceae8f1840e1381f277d9c960b3978bd542fae9f842323009ef0
Round 47: 0xad24f4f6ebf7b0a57d4977903ab3681cad345b88182bae52f8065e6147f7d8e5248bb4624477b9b3e451163a72a8ceae8f1840e1381f277d9c960b3978bd542f
Round 48: 0xb46f91dce6a4f6afad24f4f6ebf7b0a57d4977903ab3681cad345b88182bae5217006f9b018cebc6248bb4624477b9b3e451163a72a8ceae8f1840e1381f277d
Round 49: 0x0c027dba992c0952b46f91dce6a4f6afad24f4f6ebf7b0a57d4977903ab3681c119dc4afd215a05d17006f9b018cebc6248bb4624477b9b3e451163a72a8ceae
Round 50: 0x7352382665123eb10c027dba992c0952b46f91dce6a4f6afad24f4f6ebf7b0a502baea50cce1b5ea119dc4afd215a05d17006f9b018cebc6248bb4624477b9b3
Round 51: 0x066ffd750ae36bfc7352382665123eb10c027dba992c0952b46f91dce6a4f6affaa577e706a535cf02baea50cce1b5ea119dc4afd215a05d17006f9b018cebc6
Round 52: 0x6414ff1ef416602b066ffd750ae36bfc7352382665123eb10c027dba992c09520ba66fdd323032f3faa577e706a535cf02baea50cce1b5ea119dc4afd215a05d
Round 53: 0x51b8663feb69fa346414ff1ef416602b066ffd750ae36bfc7352382665123eb135d830475200b0f70ba66fdd323032f3faa577e706a535cf02baea50cce1b5ea
Round 54: 0x31d4166feffc9a8051b8663feb69fa346414ff1ef416602b066ffd750ae36bfc2d56411a0360643835d830475200b0f70ba66fdd323032f3faa577e706a535cf
Round 55: 0x265be632da920b9231d4166feffc9a8051b8663feb69fa346414ff1ef416602bdf4df5d1e4a0b2252d56411a0360643835d830475200b0f70ba66fdd323032f3
Round 56: 0xb1c1907c0661c4b4265be632da920b9231d4166feffc9a8051b8663feb69fa3469cd0989461b5ce2df4df5d1e4a0b2252d56411a0360643835d830475200b0f7
Round 57: 0xfe98939eacef5292b1c1907c0661c4b4265be632da920b9231d4166feffc9a807ef66970705de13769cd0989461b5ce2df4df5d1e4a0b2252d56411a03606438
Round 58: 0x1554b7a04b80e2a7fe98939eacef5292b1c1907c0661c4b4265be632da920b9237b818a338c80e6b7ef66970705de13769cd0989461b5ce2df4df5d1e4a0b225
Round 59: 0x80b467279aa1b2531554b7a04b80e2a7fe98939eacef5292b1c1907c0661c4b406c6a680881cfcac37b818a338c80e6b7ef66970705de13769cd0989461b5ce2
Round 60: 0x19fa1a0dbfc53eb180b467279aa1b2531554b7a04b80e2a7fe98939eacef529233a0e81aef1b0d5406c6a680881cfcac37b818a338c80e6b7ef66970705de137
Round 61: 0x9bbe6be588de39e419fa1a0dbfc53eb180b467279aa1b2531554b7a04b80e2a7ff85999dde5f075733a0e81aef1b0d5406c6a680881cfcac37b818a338c80e6b
Round 62: 0x1f69ed6e3ac993cd9bbe6be588de39e419fa1a0dbfc53eb180b467279aa1b253763a8da59746e30dff85999dde5f075733a0e81aef1b0d5406c6a680881cfcac
Round 63: 0xe51b1f63aeda0a961f69ed6e3ac993cd9bbe6be588de39e419fa1a0dbfc53eb14b765018633402a2763a8da59746e30dff85999dde5f075733a0e81aef1b0d54
Round 64: 0xd98e796fd55ed16ce51b1f63aeda0a961f69ed6e3ac993cd9bbe6be588de39e493018437250cec7d4b765018633402a2763a8da59746e30dff85999dde5f0757
Round 65: 0x0b57a0f5701a5e24d98e796fd55ed16ce51b1f63aeda0a961f69ed6e3ac993cd4afa7036c2e6efe893018437250cec7d4b765018633402a2763a8da59746e30d
Round 66: 0xa8125ab407971f3c0b57a0f5701a5e24d98e796fd55ed16ce51b1f63aeda0a961719cd444e4d10ba4afa7036c2e6efe893018437250cec7d4b765018633402a2
Round 67: 0x416d992602c8bfb7a8125ab407971f3c0b57a0f5701a5e24d98e796fd55ed16c8cd725588a1351c94719cd444e4d10ba4afa7036c2e6efe893018437250cec7d
Round 68: 0x5c29b290553c51bb416d992602c8bfb7a8125ab407971f3c0b57a0f5701a5e244c244d658000d4d88cd725588a1351c94719cd444e4d10ba4afa7036c2e6efe8
Round 69: 0x49f81d7d6b7589775c29b290553c51bb416d992602c8bfb7a8125ab407971f3ca705ac4670cab1184c244d658000d4d88cd725588a1351c94719cd444e4d10ba
Round 70: 0x36738ed1d0dc0d8249f81d7d6b7589775c29b290553c51bb416d992602c8bfb77dc93215f18a7254a705ac4670cab1184c244d658000d4d88cd725588a1351c9
Round 71: 0xfd82560b594608a336738ed1d0dc0d8249f81d7d6b7589775c29b290553c51bb081fed071bc9d4f37dc93215f18a7254a705ac4670cab1184c244d658000d4d8
Round 72: 0xf3413465494bc383fd82560b594608a336738ed1d0dc0d8249f81d7d6b7589777cf54495e4751a00081fed071bc9d4f37dc93215f18a7254a705ac4670cab118
Round 73: 0xb1462df496670ae2f3413465494bc383fd82560b594608a336738ed1d0dc0d82f77ed5d7d96901b17cf54495e4751a00081fed071bc9d4f37dc93215f18a7254
Round 74: 0x042ea4a38e8005a2b1462df496670ae2f3413465494bc383fd82560b594608a39e9a5c8988c33012f77ed5d7d96901b17cf54495e4751a00081fed071bc9d4f3
Round 75: 0xd1e51cc0a386eac0042ea4a38e8005a2b1462df496670ae2f3413465494bc383d163f2a2e6a031689e9a5c8988c33012f77ed5d7d96901b17cf54495e4751a00
Round 76: 0x04397ed950983c8ed1e51cc0a386eac0042ea4a38e8005a2b1462df496670ae2d44c6af567d2b42cd163f2a2e6a031689e9a5c8988c33012f77ed5d7d96901b1
Round 77: 0xf33b814c5335107c04397ed950983c8ed1e51cc0a386eac0042ea4a38e8005a2b14f7ac4888a2dfed44c6af567d2b42cd163f2a2e6a031689e9a5c8988c33012
Round 78: 0x4b69064eb697acf5f33b814c5335107c04397ed950983c8ed1e51cc0a386eac0bf246438faf2f637b14f7ac4888a2dfed44c6af567d2b42cd163f2a2e6a03168
Round 79: 0x7e155aab2cab3d744b69064eb697acf5f33b814c5335107c04397ed950983c8eb30f65ef6ebff3c4bf246438faf2f637b14f7ac4888a2dfed44c6af567d2b42c
Round 80: 0x59b4016b339e31697e155aab2cab3d744b69064eb697acf5f33b814c5335107c8af42f9637525ad6b30f65ef6ebff3c4bf246438faf2f637b14f7ac4888a2dfe
Hash value: 0xc3bde7d3275afa71397d0930b175e4af87d7f9c1b52ca520988b7686b252476ddc028215e538dda74e14ce7b99fe5fe3dea83de4f634b3a20d3047dd9c084f77
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Round 1: 0xc9e02e4411120284c3bde7d3275afa71397d0930b175e4af87d7f9c1b52ca520c4360e846530e6c7dc028215e538dda74e14ce7b99fe5fe3dea83de4f634b3a2
Round 2: 0x9904be1afae2d404c9e02e4411120284c3bde7d3275afa71397d0930b175e4afb998fba3d8d366f3c4360e846530e6c7dc028215e538dda74e14ce7b99fe5fe3
Round 3: 0xca8b15c987cb8d369904be1afae2d404c9e02e4411120284c3bde7d3275afa7115033fc746b5d88cb998fba3d8d366f3c4360e846530e6c7dc028215e538dda7
Round 4: 0xe6628a634a837ed0ca8b15c987cb8d369904be1afae2d404c9e02e4411120284525a8cb26695657c15033fc746b5d88cb998fba3d8d366f3c4360e846530e6c7
Round 5: 0xd2b13a46305dab10e6628a634a837ed0ca8b15c987cb8d369904be1afae2d404168c95f6782f0fc8525a8cb26695657c15033fc746b5d88cb998fba3d8d366f3
Round 6: 0xf6d47f41cd047cebd2b13a46305dab10e6628a634a837ed0ca8b15c987cb8d36c788c27a87011959168c95f6782f0fc8525a8cb26695657c15033fc746b5d88c
Round 7: 0xf41d526b7b20d13af6d47f41cd047cebd2b13a46305dab10e6628a634a837ed0a71b547573c79c6ec788c27a87011959168c95f6782f0fc8525a8cb26695657c
Round 8: 0xf1e5646702066264f41d526b7b20d13af6d47f41cd047cebd2b13a46305dab10277f0e434a4a9f71a71b547573c79c6ec788c27a87011959168c95f6782f0fc8
Round 9: 0xf5a71eeb5b5538bcf1e5646702066264f41d526b7b20d13af6d47f41cd047ceba49f30bb54e19dc1277f0e434a4a9f71a71b547573c79c6ec788c27a87011959
Round 10: 0xf9ddb35597c1c71af5a71eeb5b5538bcf1e5646702066264f41d526b7b20d13a45dc6c9aeace6c97a49f30bb54e19dc1277f0e434a4a9f71a71b547573c79c6e
Round 11: 0xfac2588e100233bdf9ddb35597c1c71af5a71eeb5b5538bcf1e5646702066264a36f0ab617dbfd145dc6c9aeace6c97a49f30bb54e19dc1277f0e434a4a9f71
Round 12: 0xb7f57f51e7da95e4fac2588e100233bdf9ddb35597c1c71af5a71eeb5b5538bc181015922d3f2a194a36f0ab617dbfd145dc6c9aeace6c97a49f30bb54e19dc1
Round 13: 0xb49489fbbe82a570b7f57f51e7da95e4fac2588e100233bdf9ddb35597c1c71a06d5df77e650063c181015922d3f2a194a36f0ab617dbfd145dc6c9aeace6c97
Round 14: 0x5945dc0d9aab1426b49489fbbe82a570b7f57f51e7da95e4fac2588e100233bdab573a69f6e5907606d5df77e650063c181015922d3f2a194a36f0ab617dbfd1
Round 15: 0x96a93b25a04fdbbb5945dc0d9aab1426b49489fbbe82a570b7f57f51e7da95e40466e515a3093db2ab573a69f6e5907606d5df77e650063c181015922d3f2a19
Round 16: 0xcc2cc19689ab831696a93b25a04fdbbb5945dc0d9aab1426b49489fbbe82a570c7edd4d9bee5d4e30466e515a3093db2ab573a69f6e5907606d5df77e650063c
Round 17: 0x4378605f2d00cac4cc2cc19689ab831696a93b25a04fdbbb5945dc0d9aab1426061d8aa606786c70c7edd4d9bee5d4e30466e515a3093db2ab573a69f6e59076
Round 18: 0x7bf5d106d0e4b3f74378605f2d00cac4cc2cc19689ab831696a93b25a04fdbbb69fcac9c12a50f62061d8aa606786c70c7edd4d9bee5d4e30466e515a3093db2
Round 19: 0xd3a200413ea1d1387bf5d106d0e4b3f74378605f2d00cac4cc2cc19689ab831666fa9f62c5f9998f69fcac9c12a50f62061d8aa606786c70c7edd4d9bee5d4e3
Round 20: 0xfa3ce3f9e84bf004d3a200413ea1d1387bf5d106d0e4b3f74378605f2d00cac44ab140e7b5e6df2366fa9f62c5f9998f69fcac9c12a50f62061d8aa606786c70
Round 21: 0x27fac1e04ceacbbafa3ce3f9e84bf004d3a200413ea1d1387bf5d106d0e4b3f76240f54eec5426a34ab140e7b5e6df2366fa9f62c5f9998f69fcac9c12a50f62
Round 22: 0x149cd87435ec364f27fac1e04ceacbbafa3ce3f9e84bf004d3a200413ea1d1387fc80abe17c7bd0f6240f54eec5426a34ab140e7b5e6df2366fa9f62c5f9998f
Round 23: 0x6949d32f65fa40be149cd87435ec364f27fac1e04ceacbbafa3ce3f9e84bf00044a5b607224ed96fa7fc80abe17c7bd0f6240f54eec5426a34ab140e7b5e6df23
Round 24: 0xc90f7d41ef0b68ac6949d32f65fa40be149cd87435ec364f27fac1e04ceacbbac562208eff4f40574a5b607224ed96fa7fc80abe17c7bd0f6240f54eec5426a3
Round 25: 0xe2eb5888741b7bb4c90f7d41ef0b68ac6949d32f65fa40be149cd87435ec364fb92dfacc5ff5d10cc562208eff4f40574a5b607224ed96fa7fc80abe17c7bd0f
Round 26: 0xb89aa9904d7a3fb9e2eb5888741b7bb4c90f7d41ef0b68ac6949d32f65fa40be9792d5a69e808538b92dfacc5ff5d10cc562208eff4f40574a5b607224ed96fa
Round 27: 0xc5ecd36a1818401db89aa9904d7a3fb9e2eb5888741b7bb4c90f7d41ef0b68ac6243ad0e7bf9feec9792d5a69e808538b92dfacc5ff5d10cc562208eff4f4057
Round 28: 0x314ae5a8626c5cb1c5ecd36a1818401db89aa9904d7a3fb9e2eb5888741b7bb445bd6806e11f6dc36243ad0e7bf9feec9792d5a69e808538b92dfacc5ff5d10c
Round 29: 0x8c32f256ab965855314ae5a8626c5cb1c5ecd36a1818401db89aa9904d7a3fb94eb964b09c9b012945bd6806e11f6dc36243ad0e7bf9feec9792d5a69e808538
Round 30: 0x363b6cd8b3a24b718c32f256ab965855314ae5a8626c5cb1c5ecd36a1818401daa93e301cbc436fa4eb964b09c9b012945bd6806e11f6dc36243ad0e7bf9feec
Round 31: 0xa0c923f25ea7f991363b6cd8b3a24b718c32f256ab965855314ae5a8626c5cb18acfb12a06c580d9aa93e301cbc436fa4eb964b09c9b012945bd6806e11f6dc3
Round 32: 0x141c62cc80a73b0aa0c923f25ea7f991363b6cd8b3a24b718c32f256ab965855507c0bd14df8b58cc8acfb12a06c580d9aa93e301cbc436fa4eb964b09c9b0129
Round 33: 0x5b7d439a936ca69c141c62cc80a73b0aa0c923f25ea7f991363b6cd8b3a24b71003ba302db72772207c0bd14df8b58cc8acfb12a06c580d9aa93e301cbc436fa
Round 34: 0x1548af3ae9e2884a5b7d439a936ca69c141c62cc80a73b0aa0c923f25ea7f99193ef1f09711f845b003ba302db72772207c0bd14df8b58cc8acfb12a06c580d9
Round 35: 0x6e0341eb96450d3b1548af3ae9e2884a5b7d439a936ca69c141c62cc80a73b0aa7ec1ea02d14138493ef1f09711f845b003ba302db72772207c0bd14df8b58cc
Round 36: 0x72b0563dd61c75716e0341eb96450d3b1548af3ae9e2884a5b7d439a936ca69cd0965a6d75f65c21a7ec1ea02d14138493ef1f09711f845b003ba302db727722
Round 37: 0xa731421c994e712472b0563dd61c75716e0341eb96450d3b1548af3ae9e2884a1cc1cbd9c12af347d0965a6d75f65c21a7ec1ea02d14138493ef1f09711f845b
Round 38: 0xc5b89d99a645e461a731421c994e712472b0563dd61c75716e0341eb96450d3be949f7b5e645e3b01cc1cbd9c12af347d0965a6d75f65c21a7ec1ea02d141384
Round 39: 0x87d34639852a142ac5b89d99a645e461a731421c994e712472b0563dd61c7571cfa27d3a9bb61c3de949f7b5e645e3b01cc1cbd9c12af347d0965a6d75f65c21
Round 40: 0x1a9714dd96772d4587d34639852a142ac5b89d99a645e461a731421c994e7124c4b491775d7619fccfa27d3a9bb61c3de949f7b5e645e3b01cc1cbd9c12af347
Round 41: 0xb93885240ce595271a9714dd96772d4587d34639852a142ac5b89d99a645e4611f12c14cb9eb3b23c4b491775d7619fccfa27d3a9bb61c3de949f7b5e645e3b0
Round 42: 0x68de2fd4b2e594d7b93885240ce595271a9714dd96772d4587d34639852a142a0d89cd18a210620e1f12c14cb9eb3b23c4b491775d7619fccfa27d3a9bb61c3d
Round 43: 0xf1a75f710bd76fe168de2fd4b2e594d7b93885240ce595271a9714dd96772d459556bc6b011723e30d89cd18a210620e1f12c14cb9eb3b23c4b491775d7619fc
Round 44: 0x28a48b03cf07f962f1a75f710bd76fe168de2fd4b2e594d7b93885240ce595272be90534b49a224e9556bc6b011723e30d89cd18a210620e1f12c14cb9eb3b23
Round 45: 0x47c363c0599b075228a48b03cf07f962f1a75f710bd76fe168de2fd4b2e594d79435720a1687c2ac2be90534b49a224e9556bc6b011723e30d89cd18a210620e
Round 46: 0xb61eaf950660824d47c363c0599b075228a48b03cf07f962f1a75f710bd76fe1adf0112a246545909435720a1687c2ac2be90534b49a224e9556bc6b011723e3
Round 47: 0x4fc7b571db3fde6cb61eaf950660824d47c363c0599b075228a48b03cf07f9620d4ba41501125b21adf0112a246545909435720a1687c2ac2be90534b49a224e
Round 48: 0xcb3e4fd337850b394fc7b571db3fde6cb61eaf950660824d47c363c0599b07528596c422743c49240d4ba41501125b21adf0112a246545909435720a1687c2ac
Round 49: 0x19760e6c7223ea8ecb3e4fd337850b394fc7b571db3fde6cb61eaf950660824d7e3bc71701dc551b8596c422743c49240d4ba41501125b21adf0112a24654590
Round 50: 0x4ce80d8c6951f14e19760e6c7223ea8ecb3e4fd337850b394fc7b571db3fde6cd13e9818ef7b202f7e3bc71701dc551b8596c422743c49240d4ba41501125b21
Round 51: 0x5f887ce79ce430104ce80d8c6951f14e19760e6c7223ea8ecb3e4fd337850b39ce38573743ff17b0d13e9818ef7b202f7e3bc71701dc551b8596c422743c4924
Round 52: 0xed4f2093c075b7c35f887ce79ce430104ce80d8c6951f14e19760e6c7223ea8e74619e18afbe1135ce38573743ff17b0d13e9818ef7b202f7e3bc71701dc551b
Round 53: 0xc46c36ee1e735f52ed4f2093c075b7c35f887ce79ce430104ce80d8c6951f14ebf531e311fa6394574619e18afbe1135ce38573743ff17b0d13e9818ef7b202f
Round 54: 0xe4406e117caa570cc46c36ee1e735f52ed4f2093c075b7c35f887ce79ce43010a746826ff73655d1bf531e311fa6394574619e18afbe1135ce38573743ff17b0
Round 55: 0x59a8590c019d33d0e4406e117caa570cc46c36ee1e735f52ed4f2093c075b7c31d6dfdbcdd2f924fa746826ff73655d1bf531e311fa6394574619e18afbe1135
Round 56: 0x43dd53f45392eba859a8590c019d33d0e4406e117caa570cc46c36ee1e735f526b0caae5caf075ea1d6dfdbcdd2f924fa746826ff73655d1bf531e311fa63945
Round 57: 0x1c3aeaa8acaefff143dd53f45392eba859a8590c019d33d0e4406e117caa570cd971a0c8fbe21a9c6b0caae5caf075ea1d6dfdbcdd2f924fa746826ff73655d1
Round 58: 0xe77a6b7c6f8bcd811c3aeaa8acaefff143dd53f45392eba859a8590c019d33d0c16560cf83fec2a8d971a0c8fbe21a9c6b0caae5caf075ea1d6dfdbcdd2f924f
Round 59: 0xf8f435da5efe2058e77a6b7c6f8bcd811c3aeaa8acaefff143dd53f45392eba8efa0e604e3598ab8c16560cf83fec2a8d971a0c8fbe21a9c6b0caae5caf075ea
Round 60: 0x96e54c883726b375f8f435da5efe2058e77a6b7c6f8bcd811c3aeaa8acaefff11167c2ab1b7af894efa0e604e3598ab8c16560cf83fec2a8d971a0c8fbe21a9c
Round 61: 0xf184ab2de5d006fe96e54c883726b375f8f435da5efe2058e77a6b7c6f8bcd81a6fe511b321014661167c2ab1b7af894efa0e604e3598ab8c16560cf83fec2a8
Round 62: 0x3c102b39ab6bf08cf184ab2de5d006fe96e54c883726b375f8f435da5efe2058b2e69ad049390a4ea6fe511b321014661167c2ab1b7af894efa0e604e3598ab8
Round 63: 0xa0dd4cefd44d91d83c102b39ab6bf08cf184ab2de5d006fe96e54c883726b375361a349eb56a6632b2e69ad049390a4ea6fe511b321014661167c2ab1b7af894
Round 64: 0xc2d9bbd49ad7dc6da0dd4cefd44d91d83c102b39ab6bf08cf184ab2de5d006fe3ac7fa4f5812d7a7361a349eb56a6632b2e69ad049390a4ea6fe511b32101466
Round 65: 0x2f1b6f605e17a577c2d9bbd49ad7dc6da0dd4cefd44d91d83c102b39ab6bf08c5cc9501122ef7ade3ac7fa4f5812d7a7361a349eb56a6632b2e69ad049390a4e
Round 66: 0xa94bea54bf7873f32f1b6f605e17a577c2d9bbd49ad7dc6da0dd4cefd44d91d80c3f1443b85eb3ff5cc9501122ef7ade3ac7fa4f5812d7a7361a349eb56a6632
Round 67: 0x305e48f1a3235642a94bea54bf7873f32f1b6f605e17a577c2d9bbd49ad7dc6db4b7d76b3f5a2a990c3f1443b85eb3ff5cc9501122ef7ade3ac7fa4f5812d7a7
Round 68: 0x32c77ca3eb8efa4f305e48f1a3235642a94bea54bf7873f32f1b6f605e17a5771289d6e8b99fbd92b4b7d76b3f5a2a990c3f1443b85eb3ff5cc9501122ef7ade
Round 69: 0xa3877e2aabc9aed432c77ca3eb8efa4f305e48f1a3235642a94bea54bf7873f39c787d04e01346191289d6e8b99fbd92b4b7d76b3f5a2a990c3f1443b85eb3ff
Round 70: 0x1a78c72535987b65a3877e2aabc9aed432c77ca3eb8efa4f305e48f1a3235624dbe1f338fb3bd369c787d04e01346191289d6e8b99fbd92b4b7d76b3f5a2a99
Round 71: 0x1125383a5ae58c001a78c72535987b65a3877e2aabc9aed432c77ca3eb8efa4fb0316a8a275935e94dbe1f338fb3bd369c787d04e01346191289d6e8b99fbd92
Round 72: 0x49a0f7fda0f2e6c01125383a5ae58c001a78c72535987b65a3877e2aabc9aed41cee8e5e3a7faaf0b0316a8a275935e94dbe1f338fb3bd369c787d04e0134619
Round 73: 0x5a732d4260a3e36049a0f7fda0f2e6c01125383a5ae58c001a78c72535987b65488683b6d288a02a1cee8e5e3a7faaf0b0316a8a275935e94dbe1f338fb3bd36
Round 74: 0x4afbcd4158b6a4685a732d4260a3e36049a0f7fda0f2e6c01125383a5ae58c00f57bcdd0bc504061488683b6d288a02a1cee8e5e3a7faaf0b0316a8a275935e9
Round 75: 0x5aa1fb9e7c17bfaa4afbcd4158b6a4685a732d4260a3e36049a0f7fda0f2e6c0c0dcc3c108b2946bf57bcdd0bc504061488683b6d288a02a1cee8e5e3a7faaf0
Round 76: 0x25da4cdab823aaec5aa1fb9e7c17bfaa4afbcd4158b6a4685a732d4260a3e360b2011d65785c654dc0dcc3c108b2946bf57bcdd0bc504061488683b6d288a02a
Round 77: 0x40bc048810872d4925da4cdab823aaec5aa1fb9e7c17bfaa4afbcd4158b6a468b6f160c36c5977bdb2011d65785c654dc0dcc3c108b2946bf57bcdd0bc504061
Round 78: 0xec845b7b4c5e3fbc40bc048810872d4925da4cdab823aaec5aa1fb9e7c17bfaada55ca85a39f0292b6f160c36c5977bdb2011d65785c654dc0dcc3c108b2946b
Round 79: 0x805d960b60689dc4ec845b7b4c5e3fbc40bc048810872d4925da4cdab823aaec95f67c7139bb83a6da55ca85a39f0292b6f160c36c5977bdb2011d65785c654d
Round 80: 0x8303ac39c9ae50bc805d960b60689dc4ec845b7b4c5e3fbc40bc048810872d491feeb90694b4ffc495f67c7139bb83a6da55ca85a39f0292b6f160c36c5977bd
Hash value: 0x46c1940cf1094b2db9da9f3c11de8273745c553d018ae4dcd9477b0ec2d974b6fbf13b1c79eddd6be40b4aecd3b9e389b8fe086a99d3b634c421a8a10861c734
Final Hash: 0x46c1940cf1094b2db9da9f3c11de8273745c553d018ae4dcd9477b0ec2d974b6fbf13b1c79eddd6be40b4aecd3b9e389b8fe086a99d3b634c421a8a10861c734
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